torch.renorm#

https://pytorch.org/docs/stable/generated/torch.renorm.html

Description:

Returns a tensor where each sub-tensor of input along dimension dim is normalized such that the p-norm of the sub-tensor is lower than the value maxnorm If the norm of a row is lower than maxnorm, the row is unchanged input (Tensor) – the input tensor. p (float) – the power for the norm computation

Function Signature

```
torch.renorm(input, p, dim, maxnorm, *, out=None) → Tensor#
Parameters
Keyword Arguments
```

Parameters

`Keyword`

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> x = torch.ones(3, 3)
>>> x[1].fill_(2)
tensor([ 2.,  2.,  2.])
>>> x[2].fill_(3)
tensor([ 3.,  3.,  3.])
>>> x
tensor([[ 1.,  1.,  1.],
        [ 2.,  2.,  2.],
        [ 3.,  3.,  3.]])
>>> torch.renorm(x, 1, 0, 5)
tensor([[ 1.0000,  1.0000,  1.0000],
        [ 1.6667,  1.6667,  1.6667],
        [ 1.6667,  1.6667,  1.6667]])
```

Notes & Warnings

NOTE Note If the norm of a row is lower than maxnorm, the row is unchanged

Detailed Documentation

Documentation

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If the norm of a row is lower than maxnorm, the row is unchanged

input (Tensor) – the input tensor.

p (float) – the power for the norm computation

dim (int) – the dimension to slice over to get the sub-tensors

maxnorm (float) – the maximum norm to keep each sub-tensor under

out (Tensor, optional) – the output tensor.

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